

Soln: p : I will go swimming q : I will stay in the sun too long.

r : I will sunburn

The form of the argument:
$$\frac{p \rightarrow q}{q \rightarrow r} \therefore p \rightarrow r$$

Rule of inference:

Hypothetical syllogism

5) Use rules of inference to show that the hypotheses "Randy works hard," "If Randy works hard, then he is a dull boy," and "If Randy is a dull boy, then he will not get the job" imply the conclusion "Randy will not get the job."

Soln: p : Randy works hard q : Randy is a dull boy r : Randy will get the job.

Premises: 1) p 2) $p \rightarrow q$ 3) $q \rightarrow \neg r$

Conclusion: $\neg r$

Steps
1) p
2) $p \rightarrow q$
3) q
4) $q \rightarrow \neg r$
5) $\neg r$

Reason

Premise

Premise

Modus ponens on 1, 2

Premise

Modus ponens on 3, 4

6) Use rules of inference to show that the hypothesis "If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on," "If the sailing race is held, then the trophy will be awarded," and "The trophy was not awarded" imply the conclusion "It rained."

Soln: p : It rained q : It is foggy r : The sailing race is held

s : The lifesaving demonstration will go on.

t : The trophy will be awarded.